

Teorema de Cauchy-Hadamard

Teorema 1 (de Cauchy-Hadamard). Sea $f(z) = \sum_n a_n(z-z_0)^n \in \mathbb{C}[[x]]$ una serie formal de potencias con coeficientes en $\mathbb{C}$, entonces

- (1) $\frac{1}{R} = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$ donde R es el radio de convergencia de la serie.
- (2) $f(z)$ converge absolutamente en $D(z_0, R) = \{z \in \mathbb{C} : |z - z_0| < R\}$.
- (3) $f(z)$ converge uniformemente en cada disco cerrado $\overline{D}(z_0, r)$ con $0 \leq r < R$.
- (4) $f(z)$ diverge en $\overline{D}(z_0, R)^c$.

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